

LESSON 2-3 • SESSION 2 • PRACTICE • FORM B

Comparing Distributions

Learning target – I can compare two distributions using the median and the interquartile range together. · 6.SP.B.5d

REMEMBER

- **Median** = center: which group is typically higher or lower?
- **IQR** = spread: which group is more **consistent** (smaller IQR)?
- Use **both** to decide. Ask what is better here: a shorter wait or a higher score?

WORKED EXAMPLE

Line 1: median 8 min, IQR 7. Line 2: median 13 min, IQR 12.

Step 1 Center: $13 - 8 = 5$, so Line 1 is typically 5 min shorter.

Step 2 Spread: $7 < 12$, so Line 1 is steadier.

Answer Join Line 1: shorter and steadier

WORDS center (*centro*) · spread (*dispersión*) · distribution (*distribución*)

PRACTICE

show your work · label units

- 1. Two Lunch Booths.** Wait times (min). Booth 1: 4, 6, 8, 9, 11, 13, 16. Booth 2: 1, 3, 10, 14, 15, 17, 21.

Hint Lower median = shorter wait. Smaller IQR = steadier.

	Median	Q1	Q3	IQR
Booth 1				
Booth 2				

Which to join _____

- 2. Two Bean Plots.** Plot A lengths (cm): 8, 10, 12, 14, 15, 17, 18. Plot B: 4, 9, 13, 14, 19, 22, 26.

a. Median A: ____ b. IQR A: ____ c. Median B: ____ d. IQR B: ____

- 3. Two Months.** Month 1: median 210 sales, IQR 15. Month 2: median 190, IQR 45.

a. Typically higher: ____

b. Steadier: ____

- 4. Rosa's Claim.** Set E: median 72, IQR 28. Set F: median 68, IQR 7. Rosa says Set E wins in every way. Is Rosa right?

Hint A higher median says nothing about spread.

☐ Right ☐ Not right

- 5. Two Fountains.** The box plots show cups served per hour.

Hint Read the box edges for Q1 and Q3.

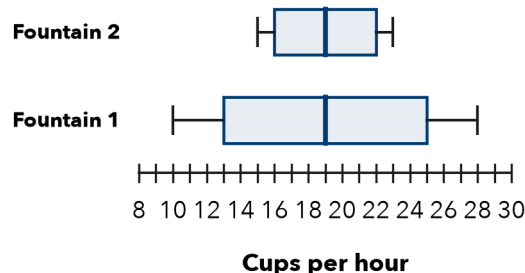
a. Median of 1: _____

b. IQR of 1: _____

c. Median of 2: _____

d. IQR of 2: _____

e. Compare them:



- 6. Coach's Choice.** Team 1: median 61 points, IQR 6. Team 2: median 66, IQR 25. The coach wants the team most likely to score close to its usual total.
- ☐ Team 1 ☐ Team 2

Why_____

- 7. Two Stores.** Store R: median 38 sales, IQR 14. Store S: median 33, IQR 5. Write a comparison that uses both measures.

- 8. Two Study Methods.** Method 1 quiz scores: median 85, IQR 20. Method 2: median 81, IQR 5.
- a.** Higher typical score: _____
 - b.** More dependable (use both IQRs): _____
 - c.** Pick one method for the whole class. Argue using both measures.

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Lucia's Work.** Lucia wrote: "Group 1 has the larger median, so Group 1 must have the larger IQR too." Explain the mistake.

TEST PRACTICE

choose the best answer

- 10.** Medians of 48 and 55 differ by –
- A** 7 **B** 103 **C** 8 **D** 55
- 11.** Group 1: median 12 cm, IQR 3 cm. Group 2: median 12 cm, IQR 9 cm. Which is more consistent?
- A** Group 1 **B** Group 2
- C** They are the same **D** Cannot tell

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

Month 1 I	Month 2 Z	18 X	Method 1 S	12 A	Booth 1 M	13 D	6 N	Booth 2 Q	7 E
#1	#2b	#2d	#3a	#5b	#5d	#8a			

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____